Basics of identification theory

Lecture 2. Approximation

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Example. Friction force identification

Friction force identification

Consider a simple model of friction force, Coulomb and viscous friction:

$$F = F_c \text{sign}(v) + F_v v, \tag{1}$$

where F_c is value of Coulomb friction coefficient, F_v is value of viscous friction coefficient and v is body velocity.

Friction force identification

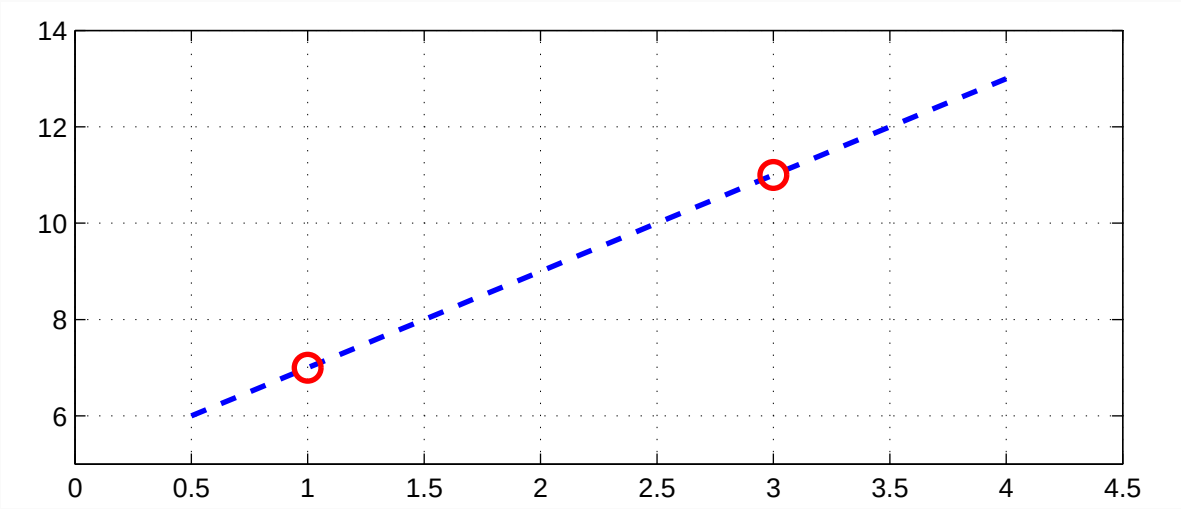
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where F_c is value of Coulomb friction coefficient, F_v is value of viscous friction coefficient and v is body velocity.
Suppose that after some experiment we have measured pairs of values (v_1, F_1) and (v_2, F_2), where $v_1, v_2 > 0$. Then

$$F_v = \frac{F_2 - F_1}{v_2 - v_1}, \tag{2}$$

$$\begin{aligned} F_c &= F_1 - F_v v_1 \\ &= F_2 - F_v v_2 \end{aligned} \tag{3}$$



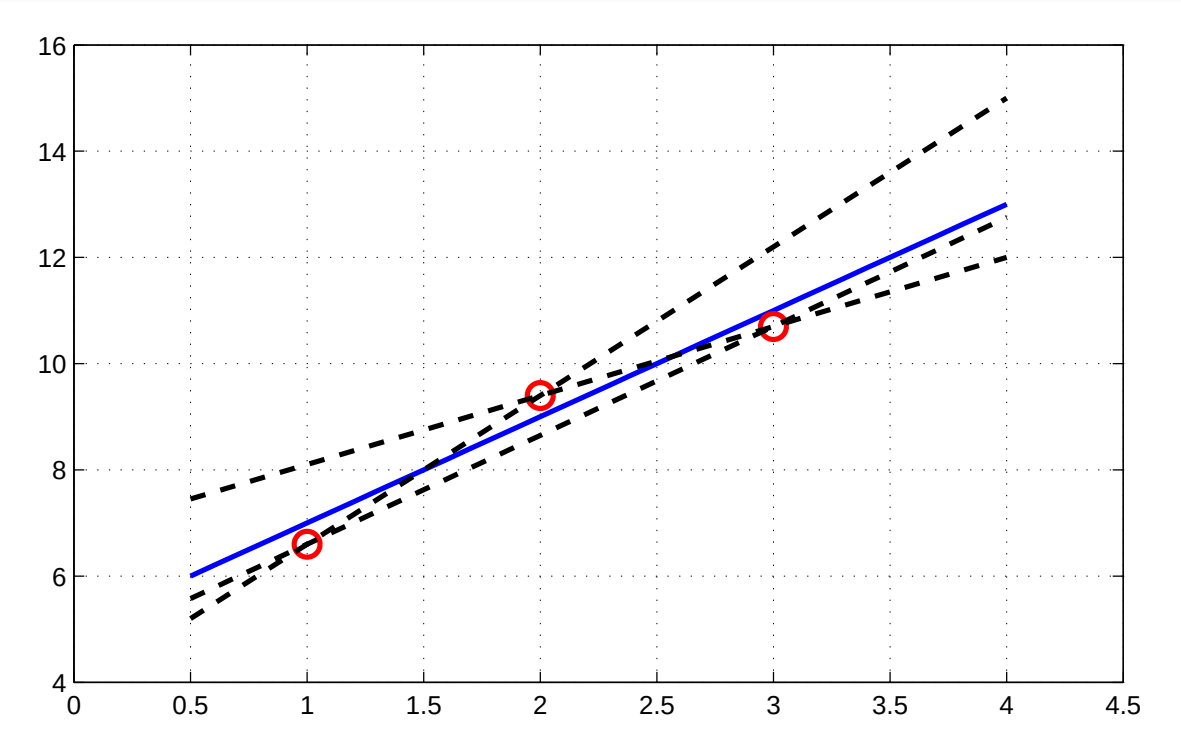
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But what if we measured three points?



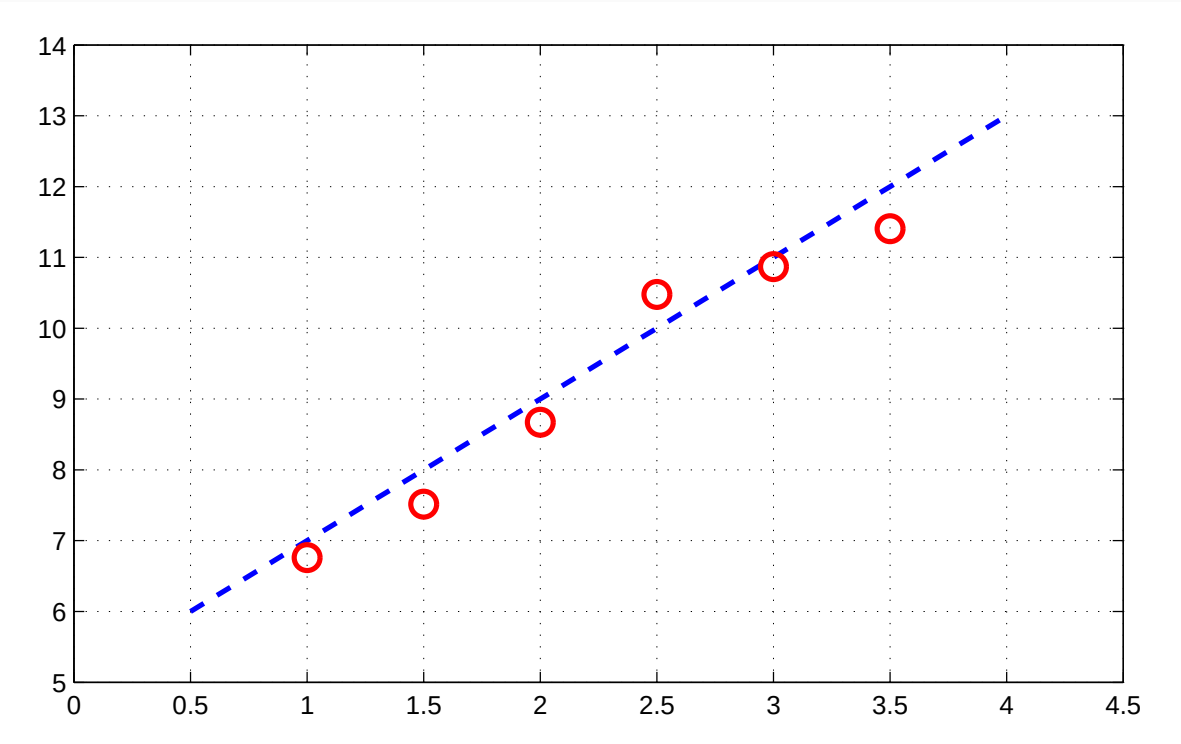
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But what if we measured three points? A lot of points?



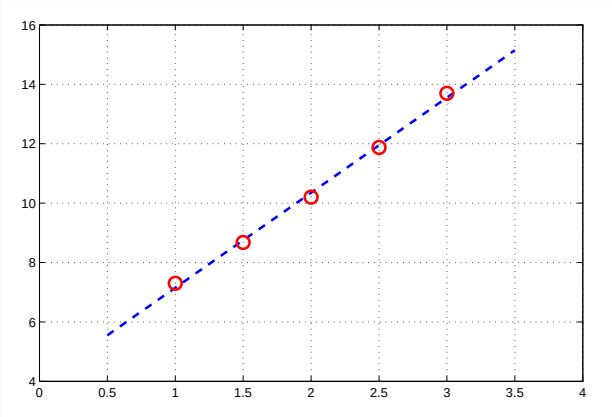
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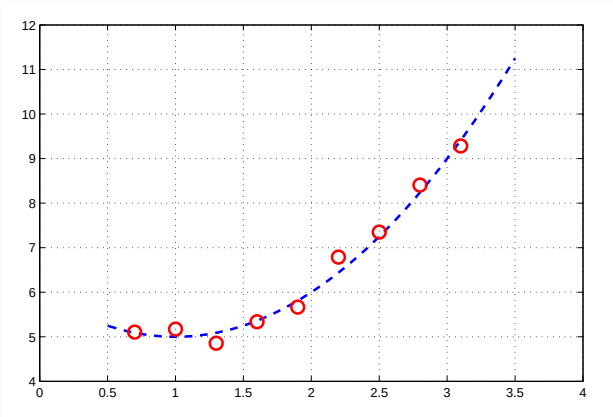
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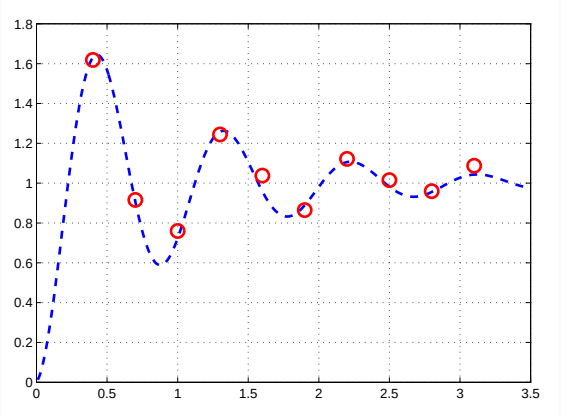
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where F_c is value of Coulomb friction coefficient, F_v is value of viscous friction coefficient and v is body velocity.

How to find the best approximation?







Least-squares method. Deterministic problem

Problem statement

Regression model: $y = g(x_1, x_2, \ldots, x_m)$, where the independent variables x_i are *regressors*, y is a *dependent variable* or *regressand*.

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Linear regression: $y = x_1\theta_1 + x_2\theta_2 + \cdots + x_m\theta_m = \sum_{i=1}^{m} x_i\theta_i$

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$$y = \mathbf{x}^T\boldsymbol{\theta}$$

where $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{x} = \text{col}\{x_i\}$ and where $\boldsymbol{\theta} \in \mathbb{R}^m$, $\boldsymbol{\theta} = \text{col}\{\theta_i\}$.

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Example

$$F = F_c\mathrm{sign}v + F_v = \begin{bmatrix}\mathrm{sign}v & v\end{bmatrix}\begin{bmatrix}F_c\\F_v\end{bmatrix} = \mathbf{x}^T\boldsymbol{\theta},$$

where $\mathbf{x} = \begin{bmatrix}\mathrm{sign}v & v\end{bmatrix}$ and $\boldsymbol{\theta} = \begin{bmatrix}F_c\\F_v\end{bmatrix}$.

Notice: that model is only an approximation.

Problem statement

Linear regression

$$y = \mathbf{x}^T \boldsymbol{\theta}$$

Suppose we obtain N measurements: $(y(k), x(k)), k = 1, \ldots, N$. Then

$$y(k) = \mathbf{x}^T(k)\boldsymbol{\theta}$$

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Linear regression

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Suppose we obtain N measurements: $(y(k), x(k)), k = 1, \ldots, N$. Then

$$y(k) = \mathbf{x}^T(k)\boldsymbol{\theta}$$

or

$$Y = X\boldsymbol{\theta}, \tag{4}$$

where $Y \in \mathbb{R}^N$, $Y = \text{col}\{y(k)\}$ and $X \in \mathbb{R}^{N \times m}$, $X = \text{col}\{\mathbf{x}^T(k)\}$,

$$X = \begin{bmatrix} x_1(1) & x_2(1) & \ldots & x_m(1) \\ x_1(2) & x_2(2) & \ldots & x_m(2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(N) & x_2(N) & \ldots & x_m(N) \end{bmatrix}$$

Problem statement

Linear regression

$$y = \mathbf{x}^T\boldsymbol{\theta} \Rightarrow Y = X\boldsymbol{\theta}$$

Problem: find the best approximation. How to evaluate the quality of the approximation?

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Problem: find the best approximation. How to evaluate the quality of the approximation?

For $\hat{\boldsymbol{\theta}}$ we can find the estimation $\hat{y}(k) = \mathbf{x}^T\hat{\boldsymbol{\theta}}$ and estimation error

$$e(k) = y(k) - \hat{y}(k) = y(k) - \mathbf{x}^T(k)\hat{\boldsymbol{\theta}},$$
$$\mathbf{e} = \mathrm{col}\{e(k)\} \in \mathbb{R}^N.$$

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Let us form the criterion (loss function) as the sum of the squares of the error

$$V(\hat{\boldsymbol{\theta}}) = \|\mathbf{e}\|_2^2 = \mathbf{e}^T\mathbf{e} = \sum_{k=1}^{N} e^2(k)$$
$$= \sum^{N}(y(k) - \mathbf{x}^T\hat{\boldsymbol{\theta}})^2 = \left(Y - X\hat{\boldsymbol{\theta}}\right)^T\left(Y - X\hat{\boldsymbol{\theta}}\right)$$

Problem statement

Linear regression

$$y = \mathbf{x}^T\boldsymbol{\theta} \Rightarrow Y = X\boldsymbol{\theta}$$

Deterministic problem

For the given Y and X find $\hat{\boldsymbol{\theta}}_{LS} = \arg\min_{\hat{\boldsymbol{\theta}}} V(\hat{\boldsymbol{\theta}})$ for

$$V(\hat{\boldsymbol{\theta}}) = \|\mathbf{e}\|_2^2 = \mathbf{e}^T\mathbf{e} = \left(Y - X\hat{\boldsymbol{\theta}}\right)^T\left(Y - X\hat{\boldsymbol{\theta}}\right)$$

Notice. In the deterministic problem formulation there is no question of identifying some true value $\boldsymbol{\theta}$, not even claim about existence of such parameters. Instead, the search for the best linear approximation of a given data set is considered. Such a problem statement is presented, for example, in statistic.

Problem statement

Linear regression

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Deterministic problem

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There is an alternative L_1 criterion

$$V(\hat{\boldsymbol{\theta}}) = \|\mathbf{e}\|_1 = \sum_{k=1}^{N} |e(k)|$$

LSM: Theorem

Deterministic problem

For the given Y and X find $\hat{\theta}_{LS} = \arg\min_{\hat{\theta}} V(\hat{\theta})$ for

$$V(\hat{\theta}) = \|\mathbf{e}\|_2^2 = \mathbf{e}^T\mathbf{e} = \left(Y - X\hat{\theta}\right)^T \left(Y - X\hat{\theta}\right)$$

Theorem (Solution of a deterministic problem)

Minimum $V(\hat{\theta}) = \left(Y - X\hat{\theta}\right)^T \left(Y - X\hat{\theta}\right)$ achieved when $\hat{\theta}$ such that

$$\left(X^TX\right)\hat{\theta} = X^TY.$$

If $\Phi \in \mathbb{R}^{m\times m}, \Phi = X^TX$ has inverse matrix then

- $\hat{\theta}_{LS} := (X^TX)^{-1}X^TY,$
- $V(\hat{\theta})$ has a single minimum:
  $\min_{\hat{\theta}} V(\hat{\theta}) = V(\hat{\theta}_{LS}) = Y^TY - Y^TX(X^TX)^{-1}X^TY.$

LSM: Proof

$$
\begin{aligned}
V(\boldsymbol{\theta}) &= (Y - X\boldsymbol{\theta})^T (Y - X\boldsymbol{\theta}) \\
&= Y^T Y - 2Y^T X\boldsymbol{\theta} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}
\end{aligned}
$$

$$
V(\boldsymbol{\theta}) \to \min \Rightarrow \Delta_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = 0
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$$V(\boldsymbol{\theta}) = (Y - X\boldsymbol{\theta})^T (Y - X\boldsymbol{\theta})$$
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$$V(\boldsymbol{\theta}) \to \min \Rightarrow \Delta_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = 0$$

Some formulas:

$$\nabla_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = \mathrm{grad}^T_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = \mathrm{col}\left\{\frac{\partial V(\boldsymbol{\theta})}{\partial \theta_i}\right\}$$

$$\nabla_{\boldsymbol{\theta}}(\mathbf{a}^T\boldsymbol{\theta}) = \Delta_{\boldsymbol{\theta}}(\boldsymbol{\theta}^T\mathbf{a}) = \mathbf{a}$$

$$\nabla_{\boldsymbol{\theta}}(\boldsymbol{\theta}^T A\boldsymbol{\theta}) = (A + A^T)\boldsymbol{\theta} = (\underbrace{\boldsymbol{\theta}^T(A + A^T)}_{\mathrm{grad}_{\boldsymbol{\theta}}})^T$$

$$X^T X = (X^T X)^T \geq 0 \ - \ \text{symmetric}$$

LSM: Proof

$$V(\boldsymbol{\theta}) = Y^T Y - 2Y^T X\boldsymbol{\theta} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}$$

$$\nabla_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = -2X^T Y + 2X^T X\boldsymbol{\theta}$$

$$\nabla_{\boldsymbol{\theta}} V(\boldsymbol{\theta}) = 0 \Leftrightarrow X^T X\boldsymbol{\theta} = X^T Y$$

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Let $X^T X$ has the inverse matrix, then

$$\boldsymbol{\theta} = \left(X^T X\right)^{-1} X^T Y =: \hat{\boldsymbol{\theta}}_{LS}$$

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and

$$\begin{aligned} V(\boldsymbol{\theta}) &= Y^T Y - 2Y^T X\boldsymbol{\theta} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta} \\ &= Y^T Y - 2Y^T X\left(X^T X\right)^{-1} X^T Y + Y^T X\left(X^T X\right)^{-1} X^T Y \\ &= Y^T Y - Y^T X\left(X^T X\right)^{-1} X^T Y \end{aligned}$$

LSM: Proof

Let us check that this is indeed the only minimum:

$$\begin{aligned} V(\hat{\theta}_{LS} + \tilde{\theta}) &= Y^T Y - 2Y^T X(\hat{\theta}_{LS} + \tilde{\theta}) + (\hat{\theta}_{LS} + \tilde{\theta})^T X^T X(\hat{\theta}_{LS} + \tilde{\theta}) \\ &= V(\hat{\theta}_{LS}) - 2Y^T X\tilde{\theta} + 2\tilde{\theta}^T X^T X\hat{\theta}_{LS} + \tilde{\theta}^T X^T X\tilde{\theta} \\ &= V(\hat{\theta}_{LS}) + \tilde{\theta}^T X^T X\tilde{\theta} \geq V(\hat{\theta}_{LS}) \end{aligned}$$

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Consider the minimum value $V(\hat{\theta}) = V(\hat{\theta}_{LS})$:

$$V(\hat{\theta}_{LS}) = Y^T Y - Y^T X(X^T X)^{-1} X^T Y. \tag{5}$$

In which case is it zero?

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$$\mathcal{I}_{N\times N} - X(X^T X)^{-1} X^T = 0.$$

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Special case: $N = m$, X - square and $\det(X) \neq 0$

Example of using the LSM for approximation

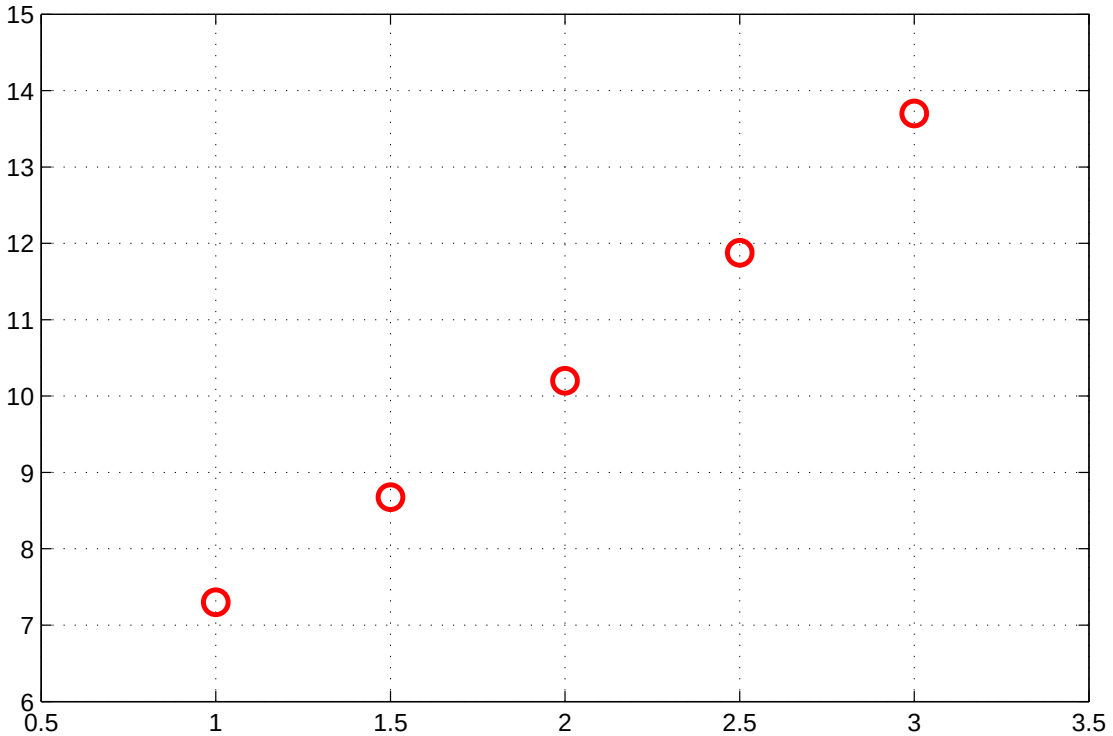
Friction

Consider the friction approximation problem with the model

$$F = F_c \text{sign} v + F_v v. \tag{6}$$

Let we have the following measurements:

v_k	F_k
1	7.3
1.5	8.7
2	10.2
2.5	11.9
3	13.7



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Consider the friction approximation problem with the model

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Let we have the following measurements:

v_k	1	1.5	2	2.5	3
F_k	7.3	8.7	10.2	11.9	13.7

In the linear regression matrix form:

$$\underbrace{\begin{bmatrix} 7.3 \\ 8.7 \\ 10.2 \\ 11.9 \\ 13.7 \end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1.5 & 1 \\ 2 & 1 \\ 2.5 & 1 \\ 3 & 1 \end{bmatrix}}_{X} \underbrace{\begin{bmatrix} F_v \\ F_c \end{bmatrix}}_{\theta}$$

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$$\begin{bmatrix} 7.3 & 8.7 & 10.2 & 11.9 & 13.7 \end{bmatrix}^T = \begin{bmatrix} 1 & 1.5 & 2 & 2.5 & 3 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}^T \begin{bmatrix} F_v \\ F_c \end{bmatrix}$$

$$X^T X = \begin{bmatrix} \sum_{k=1}^5 x_1(k)^2 & \sum_{k=1}^5 x_1(k)x_2(k) \\ \sum_{k=1}^5 x_1(k)x_2(k) & \sum_{k=1}^5 x_2(k)^2 \end{bmatrix} = \begin{bmatrix} 22.5 & 10 \\ 10 & 5 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} \sum_{k=1}^5 x_1(k)y(k) \\ \sum_{k=1}^5 x_2(k)y(k) \end{bmatrix} = \begin{bmatrix} 111.5 \\ 51.75 \end{bmatrix}$$

$$\hat{\theta}_{LS} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 3.2 \\ 3.95 \end{bmatrix}$$

Friction

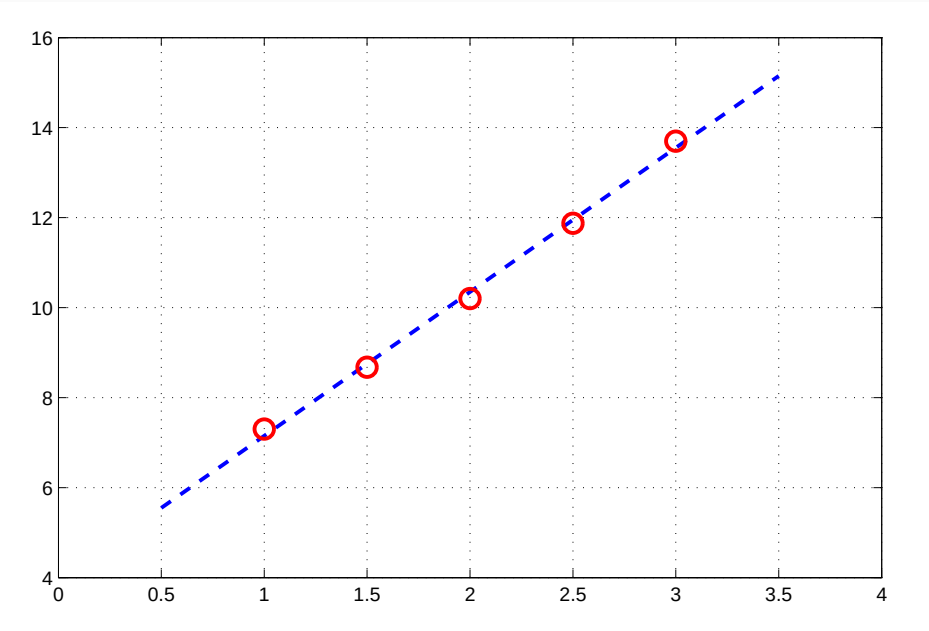
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Parabola

Consider the problem of approximating data by a parabola $y = ax^2 + bx + c$, or

$$y = \begin{bmatrix} x^2 & x & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

at N = 5 points: (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), (x_5, y_5).

Parabola

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at N = 5 points: (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), (x_5, y_5). Then:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \underbrace{\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \\ x_4^2 & x_4 & 1 \\ x_5^2 & x_5 & 1 \end{bmatrix}}_{V} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Vandermond's matrix

A matrix of the form:

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^{m-1} \\ 1 & x_2 & x_2^2 & \dots & x_2^{m-1} \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & x_3 & x_3^2 & \dots & x_3^{m-1} \end{bmatrix}$$

$V \in \mathbb{R}^{N \times m}$ is called the Vandermond matrix. For a square Vandermond matrix, the determinant

$$\det(V_{m \times m}) = \prod_{1 \le i < j \le m} (x_j - x_i),$$

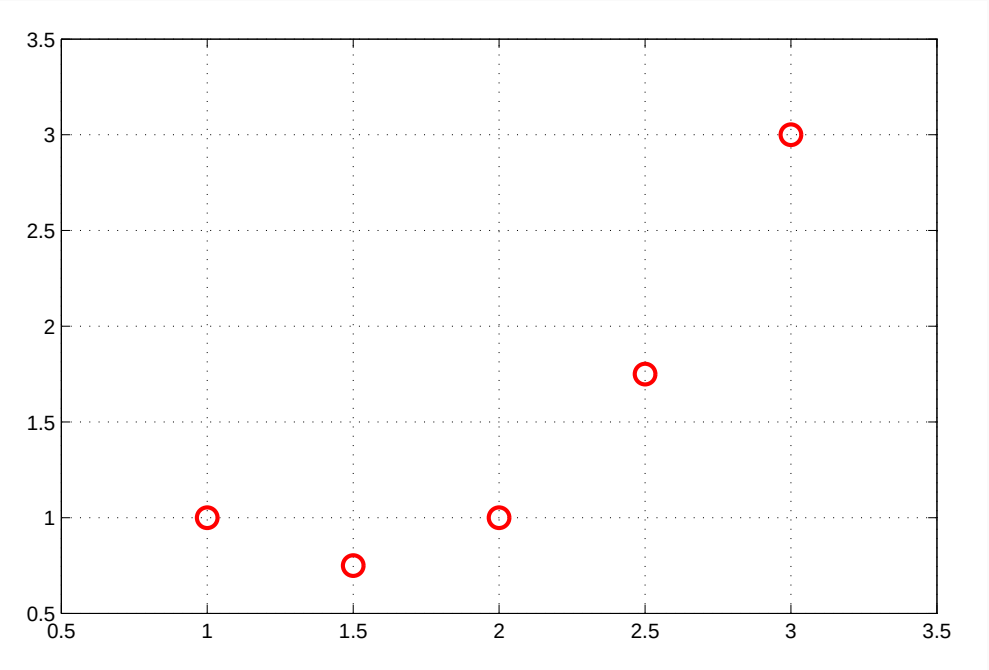
i.e., a matrix is non-degenerate if all x_i are different.
If $V \in \mathbb{R}^{N \times m}$, $N > m$, then $\det(V^T V) \neq 0$ if $\text{rank}(V) = m$, i.e., there are at least m different values of x_i.

Parabola

Consider the problem of approximating data by a parabola $y = ax^2 + bx + c$ at N = 5 points:
$(1, 1)$, $(1.5, 0.75)$, $(2, 1)$, $(2.5, 1.75)$, $(3, 3)$.

$$X^T X = \begin{bmatrix} 142.12 & 55 & 22.5 \\ 55 & 22.5 & 10 \\ 22.5 & 10 & 5 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 44.62 \\ 17.5 \\ 7.5 \end{bmatrix}$$



$$\hat{\theta}_{LS} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 1 & -3 & 3 \end{bmatrix}^T$$

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